

Speaker Notes: Lecture 19 – Full Course Review

Dr. Ronald G. Thomas

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Timing Plan

Time	Cumulative	Slide(s)	Topic
5 min	0:05	1–3	Course info, agenda
8 min	0:13	4–5	Reproducible research foundations
8 min	0:21	6–7	renv and Docker
7 min	0:28	8	Git and GitHub
7 min	0:35	9	Statistical analysis plans
7 min	0:42	10	Testing workflows
10 min	0:52	11–12	Regression and diagnostics
10 min	1:02	13–14	ML and classification metrics
10 min	1:12	15–16	Study design and cross-validation
5 min	1:17	17	Complete reproducible workflow
3 min	1:20	18	Questions

Total: approximately 80 minutes.

Slide-by-Slide Notes

Slide 1: Course Information

Standard opening. No discussion needed.

Slide 2: Lecture 19: Full Course Review

Frame the session: this is a comprehensive review of the entire quarter. Every major concept from lectures 1–18 will be revisited at a high level. Encourage students to interrupt with questions at any point – the goal is clarity, not coverage speed.

Slide 3: Today's Agenda

Walk through the ten topics. Note that these map roughly to the chronological order of the course: reproducibility foundations first, then tools, then statistical methods, then study design.

Mention that all four project datasets will be referenced throughout.

Slide 4: Reproducibility – Core Definitions

Key distinctions to emphasize:

- Reproducibility is a technical property (can I re-run your code and get the same numbers?). It is achievable with proper tools.
- Replicability is a scientific property (does the finding hold in new data?). It depends on the underlying phenomenon.
- Both matter, but reproducibility is the minimum standard.

On the five components: Ask students which component they found most difficult to maintain across their four projects. This usually generates good discussion about environment capture (renv/Docker) being the most commonly neglected.

Question discussion: The ‘different coefficients’ scenario is common in practice. Push students to think beyond ‘the code is wrong’ to structural causes. A good follow-up: ‘How would you have prevented this?’ (Answer: renv + Docker + Git.)

Anticipated student question: ‘Does reproducibility guarantee correctness?’ No. You can reproducibly produce wrong results. But reproducibility enables others to find and fix errors. Without it, errors are invisible.

Slide 5: The Reproducibility Stack

Visual emphasis: The layered diagram is the key takeaway. Each layer addresses a different failure mode:

- Code layer: the analysis itself (your scripts)
- Package layer: renv locks exact versions

- System layer: Docker locks R version + OS + system libs
- Base layer: rocker images provide stable foundations

On the tool table: Ask students to recall the three renv commands from memory. Most will get `init()` and `restore()` but forget `snapshot()` – reinforce that snapshot is the step most commonly skipped and the one that causes collaboration failures.

Question discussion: The renv insufficiency question tests understanding of the reproducibility layers. A strong answer mentions system libraries specifically. An excellent answer gives a concrete example (e.g., the sf/GDAL case, or differences in BLAS implementations affecting numerical results).

Slide 6: renv – Package Version Control

Walk through the workflow:

1. `init()` – do this once at project creation
2. Install packages as normal
3. `snapshot()` – do this after every package change
4. Commit `renv.lock` to Git
5. Collaborator clones and runs `restore()`

Emphasize the lockfile: It is JSON, human-readable, and includes hashes for integrity verification. Students should know what it contains conceptually even if they do not read it.

Question discussion: The glmnet scenario tests whether students understand the snapshot step. Common mistake: installing packages and committing code without updating the lockfile. The `renv::status()` command is a useful check they should run before every commit.

Anticipated student question: ‘What if `renv::restore()` fails because a package was removed from CRAN?’ Point to the CRAN archive, GitHub sources, and `renv::install("pkg@version")`.

Slide 7: Docker – Environment Containerization

On the Dockerfile: Walk through each line and explain what it does. Emphasize the ordering – renv files before analysis files – and explain Docker layer caching.

Key flags to reinforce:

- `--rm`: container is temporary, removed after exit
- `-it`: interactive terminal (needed for R console)
- `-v`: volume mount (your local files inside the container)

Question discussion: The layer caching question tests understanding of Docker’s build process. This is a practical optimization that matters during development. Students who built Docker images for Project 2 likely experienced slow rebuilds and will appreciate this explanation.

Anticipated student question: ‘Do I need Docker for every project?’ Not necessarily. For coursework, `renv` + `Git` is often sufficient. Docker becomes essential when sharing with collaborators on different operating systems or when deploying to servers.

Slide 8: Git and GitHub

Core workflow: Write the four commands on the board if possible. The staging area concept is the most confusing for new Git users – emphasize that staging lets you select which changes to include in a commit.

Collaboration workflow: This is the workflow students used for all four projects. Ask if any teams encountered merge conflicts and how they resolved them.

Question discussion: The merge conflict scenario tests practical Git skills. Walk through the conflict markers carefully. Emphasize the importance of communication – never resolve a conflict in analysis code unilaterally. The wrong merge could silently change results.

Anticipated student question: ‘What if I accidentally committed sensitive data?’ Use `git filter-branch` or BFG Repo-Cleaner, but emphasize prevention (`.gitignore`) over cure.

Slide 9: Statistical Analysis Plans

Key message: The SAP is not bureaucratic overhead. It is the most important document in a biostatistical project because it separates planned analyses from post-hoc exploration.

Table shells: This is often the most surprising concept for students. Show how a blank Table 1 shell forces you to identify all variables, stratification groups, and summary statistics before touching data.

Question discussion: The table shell question connects to clinical trial practice (ICH E9). For observational studies, table shells are equally valuable for preventing ‘fishing expeditions.’ A student who has worked on Project 4 (the RCT) should be able to explain this from experience.

Slide 10: Testing Data Analysis Workflows

Taxonomy: Distinguish the four test types clearly. Unit tests are for individual functions. Data validation tests check data quality (ranges, types, missing patterns). Integration tests verify the full pipeline. Reproducibility tests confirm determinism.

Code example: Walk through the `clean_penguins` test. Highlight the Arrange-Act-Assert pattern. Note that the test uses synthetic data with known properties, not the real dataset.

Question discussion: The `expect_equal` vs `expect_identical` distinction is subtle but important. Many students will have encountered floating-point comparison issues in their projects. The tolerance parameter in `expect_equal` is the solution.

Anticipated student question: ‘How many tests should I write?’ Enough to catch the errors that matter. Focus on data loading, cleaning functions, and model outputs. Testing every line of exploratory code is unnecessary.

Slide 11: Linear Regression – Simple to Species-Aware

Sequential model building: This is the pedagogical strategy from Lecture 10. Each model adds complexity and shows how R-squared improves. The jump from 0.779 to 0.868 when adding species demonstrates how much variance is explained by group membership.

Model comparison criteria (new for Test 2): Walk through all four criteria explicitly. These map directly to Test 2, Q2:

1. Adjusted R-squared – penalizes adding weak predictors (unlike raw R-squared, which can only increase)
2. RMSE – interpretable in original units (grams for penguins)
3. AIC/BIC – information-theoretic; lower is better. Note that BIC penalizes complexity more heavily than AIC, so BIC tends to select simpler models.
4. F-test for nested models – `anova()` in R. Only valid when comparing models where one is a strict subset of the other.

Show the `anova()` and `AIC()` code. Ask students: ‘Can you use an F-test to compare the species-aware model to a completely different model with different predictors?’ (No – models must be nested.)

Coefficient interpretation: Emphasize that the intercept of -5781 g is extrapolation (no penguin has zero flipper length) and should not be interpreted literally.

Question discussion: The Simpson’s Paradox question is central to the entire penguins analysis. Draw the two scatter plots on the board if possible: overall negative trend, within-species positive trend. Ask students to identify the confounding variable (species). This concept applies directly to Projects 2 and 3 as well.

Slide 12: Model Diagnostics and Cross-Validation

Diagnostic plots: Name all four and state what each reveals. The residuals vs. fitted plot is the most informative single diagnostic. Random scatter = good. Patterns = problems.

Six assumptions (new for Test 2): The slides now list all six regression assumptions from the analysis plan diagnostic table. The addition of independence (item 5) is important – emphasize that this is not checkable from a plot. It depends on knowing the data collection process (e.g., are penguins sampled independently, or are there nest-level correlations?).

VIF: Students should know the 5/10 thresholds. For the penguins data, VIF is moderate because bill measurements and flipper length are correlated.

Question discussion: The overfitting scenario (training R-squared 0.92, CV R-squared 0.71) is a critical concept. Ask students if they encountered this in Project 2 (ADNI). Many will have seen this gap with complex ML models. The remedies (regularization, feature selection, more data) are practical and important.

Slide 13: Predicting MCI to AD Conversion

Clinical context: Spend a moment on why this prediction task matters clinically. Early identification of converters enables enrollment in clinical trials for disease-modifying therapies. This is not an abstract ML exercise.

Grassi et al. approach: Emphasize the deliberate restriction to baseline clinical features. This is a design decision, not a limitation. The AUROC of 0.88 with only clinical features is clinically meaningful.

Question discussion: The biomarker exclusion question tests whether students understand the difference between maximizing accuracy in a research setting and building a deployable clinical tool. A model requiring PET scans excludes most clinical settings worldwide. This connects to the broader theme of designing analyses for real-world impact.

Slide 14: Evaluation Metrics for Classification

Metrics table: Walk through each metric with the ADNI example. Sensitivity and specificity are conditional on true status. PPV and NPV are conditional on test result. These conditioning directions are frequently confused.

ROC curve: Explain that the curve shows all possible sensitivity/specificity tradeoffs. The ‘best’ threshold depends on the clinical context – is it worse to miss a converter or to falsely alarm a stable patient?

Youden’s J statistic (new for Test 2): The formula is $J = \text{Sensitivity} + \text{Specificity} - 1$. Geometrically, it finds the point on the ROC curve farthest from the diagonal. Walk through

why this is a reasonable default: it equally weights sensitivity and specificity. Note that `coords(roc_obj, "best")` in pROC uses Youden's J by default. Ask students: 'When might you NOT want to use Youden's J?' (When the costs of false positives and false negatives are very different – e.g., a screening test for a fatal disease should favor sensitivity over specificity.)

Question discussion: The 'always predict stable' scenario is the single most important lesson about class imbalance. 75% accuracy sounds good but is useless. This directly demonstrates why accuracy alone is insufficient and why AUROC is the standard metric for binary classification.

Anticipated student question: 'How do you choose the classification threshold?' It depends on the cost of false positives vs. false negatives. For screening (catching all converters), prioritize sensitivity. For confirmatory testing (avoiding unnecessary interventions), prioritize specificity.

Slide 15: Four Datasets, Four Design Contexts

Project comparison table: This is a synthesis slide. Walk through each row and ask students to name the key analytical challenge they encountered.

Common thread: Emphasize that despite different designs (observational, cohort, RCT), the reproducible workflow was identical across all projects: SAP, compendium, renv, Git, tests, report.

Question discussion: The RCT vs. observational distinction is fundamental to biostatistics. Randomization is the key word. Ask students: 'Could we randomize officers to stop drivers of different races?' (No – ethical and practical impossibility.) This is why the policing analysis cannot make causal claims despite having millions of observations.

Slide 16: Cross-Validation Strategies

Strategy table: Each row connects to a specific project. k-fold for Penguins, site-independent for ADNI, stratified for any imbalanced outcome.

Question discussion: The site-independent split question is the most important validation concept from Project 2. Site-level correlations are a form of data leakage – subjects within a site are not independent. This applies broadly to any multicenter clinical study.

Anticipated student question: 'Is leave-one-out CV ever better than k-fold?' LOOCV has lower bias but higher variance than k-fold. For most biostatistics applications with reasonable sample sizes, 10-fold CV is preferred.

Slide 17: The Complete Reproducible Workflow

Seven steps: This is the capstone slide. Walk through the workflow as a linear process, but note that in practice it is iterative – you often loop back to earlier steps.

Question discussion: The mixed-effects model change scenario tests whether students can trace a methodological change through all workflow steps. The correct answer touches SAP, renv, code, tests, documentation, Git, and Docker. This comprehensive thinking is the goal of the entire course.

Slide 18: Questions

Open the floor. If no questions arise, consider asking:

- ‘Which tool or concept from this course do you think will be most valuable in your career?’
- ‘What would you do differently if you started Project 1 over again today?’
- ‘Is there anything from the course that still feels unclear?’

Troubleshooting Guide

If students ask about specific test format: Redirect to the test itself on Thursday. This lecture is a conceptual review, not a test prep session.

If discussion runs long on one topic: The reproducibility and regression sections are the most likely to generate extended discussion. If running behind, abbreviate the cross-validation strategies slide and direct students to the Lecture 10 slides for detailed review.

If students request practice problems: Suggest they re-read the Q&A fragments on each slide, as these are representative of the level of reasoning expected.

Prerequisites Verified in This Lecture

Students should be able to:

- ☐ Define reproducibility, replicability, and transparency
- ☐ Explain the roles of renv, Docker, Git, and rrtools
- ☐ Describe the core renv workflow (init, snapshot, restore)
- ☐ Read a Dockerfile and explain layer caching
- ☐ Execute the Git staging-commit-push workflow
- ☐ Resolve a merge conflict
- ☐ List the sections of a statistical analysis plan
- ☐ Write a testthat test with appropriate expectations
- ☐ Interpret regression coefficients and R-squared

- ☐ List the four model comparison criteria (Adj. R-sq, RMSE, AIC/BIC, F-test)
- ☐ Name six regression assumptions and their diagnostic methods
- ☐ Explain Simpson's Paradox with the penguins example
- ☐ Define Youden's J statistic and its purpose
- ☐ Distinguish sensitivity, specificity, PPV, NPV, and AUROC
- ☐ Explain why accuracy is misleading for imbalanced outcomes
- ☐ Compare RCT and observational study designs
- ☐ Justify site-independent cross-validation for ADNI
- ☐ Trace a methodological change through the full workflow